MSML606
Project 2 Proposal

Shortest path on Manhattan’s road network

Group members:

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Problem statement:

Road networks are massive, complex interconnections of paths that lots of real-world actors have to navigate daily. From emergency services to food deliveries, a brute-force approach isn't a feasible way to find the shortest route from point A to point B. This program uses a real road network obtained from OpenStreetMap via OSMnx library, specifically the road network of Manhattan, New York, and implements Dijkstra's algorithm to show how it can be used to find the shortest path between two points selected by the user in a city's road network.

Algorithm description:

Like any greedy algorithm, Dijkstra’s makes the next best move toward finding a solution, given the state it got to by making the previous move. What differentiates it from others is the fact that it works on weighted graphs and maintains a priority queue of each unvisited node, where the priority of the nodes depends on their current known shortest path from the source. Each step selects (pops) the unvisited node from the priority queue with the shortest path from the source node. If a shorter path is found through the selected node, the algorithm updates the distance of its neighbors (in the priority queue). The shortest path is guaranteed as long as the edge weights are non-negative.

Algorithm usage:

The user will select a couple of points, source and destination, from the road network map. Subsequently, Dijkstra’s algorithm will run starting from the source, and the map will show the animation of the algorithm’s exploration until it reaches its destination. Upon destination arrival, the shortest path is highlighted in a different color. When a user clicks a point in the map, the program will snap the click to the closest node, marking it as a source/destination. Each road’s intersection will be represented as a node, and each road will be represented as an edge, with its distance in meters being the edge’s weight.

AI Usage Statement: We did not use AI tools in the making of this proposal